

How Low Can Chronos-2 Go? Post-Training Quantization of a Time-Series Foundation Model

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Abstract

Time-series foundation models (TSFMs) such as Chronos-2 deliver strong zero-shot probabilistic forecasts, but their deployment cost is rarely studied and, unlike large language models, *nothing* is known about how they respond to post-training quantization (PTQ). We present the first systematic PTQ study of TSFMs: 39 weight-quantization configurations spanning seven method families (round-to-nearest, torchao, bitsandbytes INT8/NF4, HQQ, GPTQ, GPTAQ, and activation-aware scale folding) and two- to eight-bit precision, evaluated zero-shot on the 27-dataset Chronos benchmark across four models and three architectures (encoder-only Chronos-2, encoder-decoder Chronos-Bolt, decoder-only TimesFM-2.5). Beyond point (MASE) and probabilistic (weighted quantile loss, WQL) accuracy, we introduce two calibration-aware diagnostics—mean absolute coverage error and a quantile-crossing rate—because a probabilistic forecaster can degrade in ways aggregate accuracy cannot see. Five findings emerge. (i) 8-bit quantization is free for every method. (ii) Hessian-aware GPTQ retains full-precision accuracy down to **3 bits** (WQL retention 1.05), and asymmetric-calibration GPTAQ extends the parity frontier (1.016); every PTQ method collapses at 2 bits. (iii) The *method* matters far more than the *bit-width*: at 4 bits, retention spans 0.996–1.66. (iv) Interval calibration and quantile monotonicity degrade *before* accuracy does—and we trace this damage *entirely* to the 3-linear quantile head: protecting 3.2% of the weights eliminates it at every bit-width, whereas accuracy damage is distributed across the encoder. (v) Two cheap repairs follow: post-hoc quantile sorting (free, provably never worse) eliminates crossing, and split-conformal

recalibration restores nominal coverage for *every* method at *every* bit-width—even fp32 improves. We release the framework, which reproduces published per-task benchmark numbers to $< 10^{-6}$, and all per-series predictions.

1. Introduction

Time-series foundation models (TSFMs)—Chronos (Ansari et al., 2024), Chronos-2 (Ansari et al., 2025), TimesFM, Moirai—forecast unseen series zero-shot and are increasingly deployed on CPU-only servers, edge devices, and high-throughput batch pipelines where full-precision inference is costly. Quantization is the standard efficiency lever for large language models (LLMs), but its lessons do not obviously transfer to TSFMs, which are small (Chronos-2 has 120M parameters, a regime where quantization bites harder (Li et al., 2024)), encoder-only and non-autoregressive (no KV cache, so LLM decode-centric methods do not apply), and—critically—emit *probabilistic* forecasts whose calibration and quantile monotonicity matter, not just point accuracy.

Despite an extensive LLM-quantization literature, **no published work quantizes the weights of any Chronos-family or comparable TSFM** (in that literature “quantization” usually denotes Chronos-1’s input value-*binning*, a distinct notion). We close that gap with a controlled study across method families, bit-widths, and models, and argue that evaluating a *quantized probabilistic* forecaster demands metrics beyond those on TSFM leaderboards.

Contributions.

- The first systematic PTQ study of TSFMs: 39 configurations across 7 method families, 5 bit-widths, and 4 models spanning 3 architectures (Chronos-2, Chronos-Bolt small/base, TimesFM-2.5), zero-shot on 27 datasets with bootstrap confidence intervals on every aggregate.
- A calibration-aware evaluation protocol for compressed probabilistic forecasters, adding interval-coverage error and a quantile-crossing rate, which reveal failure modes invisible to WQL/MASE.
- A *mechanism*: module-family sensitivity and weight-

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space SNR diagnostics showing calibration damage is gated exclusively by the 3-linear quantile head, while accuracy damage is distributed.

- Two mitigations with a deployment recipe: an fp32 head (prevention, 3.2% of weights) and post-hoc sorting + split-conformal recalibration (cure, at any bit-width).
- An open framework that reproduces published benchmark numbers to floating-point precision and releases all per-series predictions.

2. Setup

Models. Chronos-2 (Ansari et al., 2025) (amazon/chronos-2) is a 119.5M-parameter, T5-derived *encoder-only* transformer (12 layers, $d_{\text{model}}=768$, RoPE, alternating “time” and “group” attention). It ingests patches of a per-instance-normalized series through a residual embedding (no input tokenization) and emits, in one non-autoregressive forward pass, direct multi-step forecasts at 21 quantile levels. It ships in fp32 (478 MB). Quantization targets all 126 nn.Linear weights—including the patch embeddings and the 3-linear quantile head (3.2% of weights), except in arms that explicitly protect the head; LayerNorm stays in fp32, and bf16 (not fp16, which overflows in T5-lineage models) is the deployment baseline. As cross-model arms we also quantize Chronos-Bolt small/base (Ansari et al., 2024) (48M/205M), *encoder-decoder* T5 forecasters, through the same harness (fp32 reproduces their published per-task numbers to 0.34%/0.72%; their quantile head is protected by default), and TimesFM-2.5 (google/timesfm-2.5-200m-pytorch, 200M), a *decoder-only* autoregressive forecaster with a continuous quantile head, via a torch-only adapter (deterministic single decode step at horizons ≤ 56 ; head and tokenizer protected as in our “+head” arms). TimesFM-2.5 has no published per-task reference on this suite (we sanity-anchor the adapter by its dev-subset skill exceeding Chronos-2’s); its built-in `fix_quantile_crossing` auto-sort is disabled in all arms so QCR is measured honestly.

Methods. We evaluate weight-only PTQ across seven families runnable on commodity hardware: **RTN** (symmetric round-to-nearest, per-channel/per-tensor, plus a simulated sub-8-bit sweep); **torchao** INT8 weight-only and INT8 dynamic-activation (W8A8); **bitsandbytes** LLM.int8() (Dettmers et al., 2022) and NF4 (Dettmers et al., 2023); **HQQ** (Badri & Shao, 2023), a calibration-free half-quadratic solver; **GPTQ** (Frantar et al., 2023), a Hessian-aware error-compensating quantizer implemented directly on the model’s linear layers with one-shot calibration; **GP-TAQ** (Li et al., 2025), its asymmetric-calibration extension that corrects each layer toward the *full-precision* network’s activations; and **AWQ-style** activation-aware scale folding

(Lin et al., 2024). Methods are exercised at W8, W4 (group size 64/128), W3, and W2—39 full-benchmark configurations in total.

Protocol & metrics. We evaluate zero-shot on the 27-dataset Chronos benchmark (Ansari et al., 2024) (190,674 forecasts) using the `fev` library—the harness the Chronos-2 authors use—so our numbers are directly comparable with the public leaderboard. Point accuracy is MASE and probabilistic accuracy is weighted quantile loss (WQL, 9 levels); our full-precision Chronos-2 attains 0.425 WQL skill and an 88% win rate over seasonal-naive, and our seasonal-naive matches the published per-task MASE/WQL to $< 10^{-6}$. The key quantity is **retention**: the geometric-mean ratio $\text{WQL}(\text{quant})/\text{WQL}(\text{fp32})$ over the 27 tasks (1.00 = parity), with a 1000-resample bootstrap 95% CI. Because a probabilistic forecaster can lose quality without moving WQL, we add two diagnostics. **MACE** (mean absolute coverage error) averages $|\text{empirical} - \text{nominal}|$ over the central 80/60/40/20% intervals. **QCR** (quantile-crossing rate) is the fraction of forecast points whose predicted quantiles are non-monotone. Two further protocols: a *module-family sensitivity* sweep (dev tier: 8-task subset) that partitions the 126 linears into six disjoint families, quantized or protected one family at a time; and an offline *repair* arm applying monotone rearrangement (quantile sorting) (Chernozhukov et al., 2010; Fakoor et al., 2023) and 2-fold series-split conformal recalibration (Romano et al., 2019) to the persisted predictions of every full-tier arm.

3. Results

Table 1 summarizes representative configurations; Fig. 1 plots the accuracy-bit curve and calibration. The complete 39-variant matrix is released.

F1—8-bit is free, for every method. All six W8 configurations retain WQL within [0.999, 1.013] and MASE within [1.000, 1.011] of fp32 (Table 1; Fig. 1a), at $4\times$ compression. Even W8A8 with dynamic-activation quantization costs only 1.3%. Eight-bit quantization of Chronos-2 is, for practical purposes, lossless regardless of method.

F2—error-compensating PTQ reaches full precision at 3 bits; everything collapses at 2. GPTQ retains parity at 4 bits (retention 1.00, both group sizes) and at 3 bits (1.047, CI [0.988, 1.137])—statistically indistinguishable from fp32, where HQQ loses 23% and naive RTN 70% (Fig. 1a). GPTAQ extends the frontier: W3 retention 1.016 [0.988, 1.053] and W4 parity (1.000). Its asymmetry strength matters: the official $\alpha=0.25$ beats $\alpha=1.0$ (1.016 vs 1.087 at W3). At 2 bits every PTQ method collapses—GPTAQ 1.94, GPTQ 2.17, RTN-sim 4.08—so sub-3-bit deployment needs quantization-aware training (future work). (A dev-subset signal that GPTQ *beat* fp32 resolved to parity

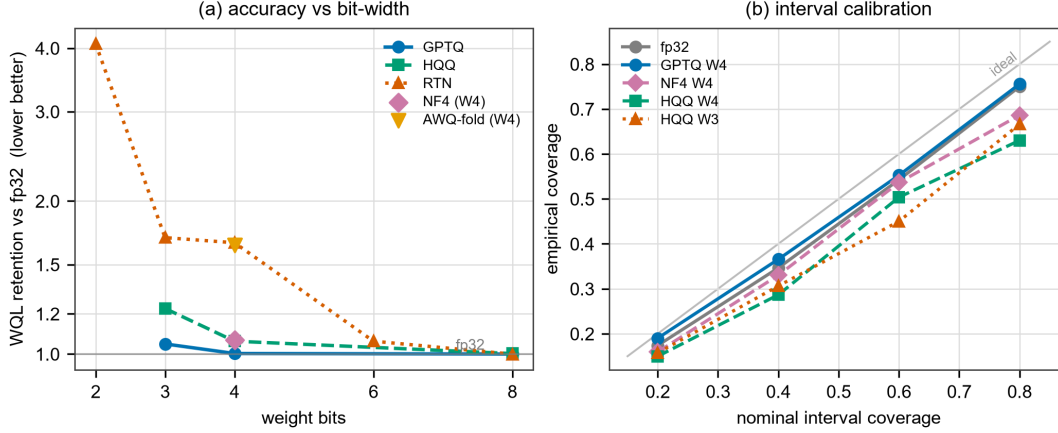


Figure 1. (a) WQL retention vs weight bits by method (log scale; lower is better, 1.0 = fp32). GPTQ tracks fp32 to 3 bits; the method spread at 4 bits exceeds the 8→4-bit gap of any single method; AWQ-fold sits on the RTN line. (b) Empirical vs nominal interval coverage: GPTQ-W4 tracks fp32 while HQQ/NF4 systematically under-cover.

Table 1. Representative variants, full 27-task benchmark. Retention = geometric-mean WQL(quant)/WQL(fp32), [95% bootstrap CI]. c80 = empirical coverage of the nominal-80% interval (fp32: 0.750). QCR = quantile-crossing rate (fp32: 0.039). BPW = effective bits/weight incl. scales; † accuracy-exact simulation (de-quantized grid in fp32)—BPW is the *achievable* footprint. “+head” = all linears quantized except the 3-linear quantile head (kept fp32).

Variant	Bits	BPW	WQL ret. [CI]	MASE	c80	QCR
fp32 (ref)	32	32.0	1.000	1.000	.750	.039
RTN W8	8	8.03	0.999 [0.997,1.00]	1.001	.749	.149
GPTQ W8	8	8.13†	1.000 [0.990,1.01]	1.003	.749	.064
HQQ W8	8	9.00	1.002 [1.00,1.00]	1.000	.746	.086
RTN W8+head	8	8.03	0.999 [0.998,1.00]	1.000	.751	.037
GPTQ W4	4	4.13†	1.002 [0.985,1.02]	1.011	.756	.254
GPTAQ W4	4	4.25†	1.000 [0.987,1.01]	1.008	.745	.226
HQQ W4	4	5.00	1.060 [1.02,1.12]	1.027	.630	.420
NF4 W4	4	4.13	1.065 [1.00,1.15]	1.026	.686	.396
HQQ W4+head	4	5.00	1.020 [0.985,1.06]	1.016	.751	.021
NF4 W4+head	4	4.13	1.045 [0.984,1.13]	1.025	.761	.022
AWQ W4	4	4.03†	1.639 [1.38,2.01]	1.409	.481	.599
GPTQ W3	3	3.25†	1.047 [0.988,1.14]	1.059	.791	.366
GPTAQ W3	3	3.25†	1.016 [0.988,1.05]	1.034	.770	.379
HQQ W3	3	4.20	1.229 [1.09,1.48]	1.143	.666	.589
GPTAQ W2	2	2.25†	1.937 [1.55,2.48]	1.816	.642	.833
RTN W2	2	2.0†	4.082 [3.08,5.57]	3.108	.060	.523

at full scale; we report the confirmed result.)

F3—method dominates bit-width. At a fixed 4 bits, WQL retention spans 0.996 (GPTQ) to 1.66 (RTN)—a 66-point gap—whereas moving a *good* method from 8 to 4 bits costs ≤6% (Fig. 1a). For a TSFM, *how* you quantize matters far more than *how much*, inverting the intuition that bit-width is the primary knob.

F4—calibration degrades before accuracy, and the quantile head is the entire mechanism. At 8 bits, accuracy is perfectly retained yet the quantile-crossing rate already quadruples (0.039 → 0.149 for RTN)—a defect WQL and MASE cannot see; at 4 bits HQQ and NF4 sag to c80 0.630/0.686 (a nominal “80%” band that holds 63%) while GPTQ keeps 0.756. Sensitivity experiments explain *why*. Partitioning the 126 linears into disjoint families (dev tier, W4 RTN-sim): quantizing *only* the 3-linear quantile head—3.2% of weights—reproduces the full collapse (WQL 2.115, vs 2.130 with all 126 quantized), while protecting only the head recovers to 1.143 with *better-than-fp32* calibration (c80 0.778, QCR 0.002). Accuracy damage is instead distributed (only-group-attention 1.001, only-FFN 1.086), and first/last-block-protection folklore does not hold here. Full-benchmark confirmation (Table 1, “+head”): an fp32 head returns W8-RTN’s QCR to 0.037 (fp32: 0.039)—the F4 quadrupling is *entirely* head-gated—and repairs W4 coverage: HQQ 0.630→0.751 (WQL 1.020), NF4 0.686→0.761 (1.045), at a cost of ≈11–13 MB. Accuracy-only evaluation therefore overstates a compressed forecaster’s quality (Dettmers et al., 2024)—but prevention costs 3% of the weight budget.

F5—damage is tail-concentrated. Mean retention hides worst cases: HQQ-W3 averages 1.23 but its worst task is 8.2× and 44% of tasks exceed 1.10; GPTQ-W4 has a median per-task ratio of 0.998 and only 4% of tasks above 1.10. Per-series distributions—released for every run—are the honest picture.

F6—no LLM-style outliers: the head fails on weight-space SNR, explaining the AWQ negative result. For Chronos-2 the AWQ scale fold is *exact* (we verify the fold numerically)—yet AWQ-fold+RTN at 4 bits (1.64) barely

differs from plain RTN (1.66) and is far behind GPTQ. Activation diagnostics show why: Chronos-2 has *no* massive-magnitude activation outliers (global $|x|_{\max} \approx 11$; the fraction of channels exceeding $|x| > 6$ is zero in most layers and at most 1.6%), only ReLU-sparsity channel heterogeneity (per-channel kurtosis is modest; pooled kurtosis is inflated by dead channels). The head’s failure is instead a weight-space signal-to-noise problem: at W4 its output layer’s relative reconstruction error is 7.83—quantization noise $\approx 8\times$ the output signal, over $100\times$ any other linear (all < 0.07)—because small quantile differences ride on large weights with no downstream layer to compensate. Activation *scaling* cannot fix a weight-space SNR deficit; Hessian-aware *compensation* (GPTQ/GPTAQ) can.

F7—post-hoc repairs: sorting is free; conformal restores coverage everywhere. Sorting predicted quantiles (monotone rearrangement) provably never increases quantile loss (Chernozhukov et al., 2010; Fakoor et al., 2023); empirically it zeroes QCR for *every* arm at never-worse WQL, and even recovers accuracy where crossing is severe (HQQ-W3 1.229 $\rightarrow$ 1.170). Coverage is only partially restored (HQQ-W4 c80 0.630 $\rightarrow$ 0.703): crossing is cosmetic damage; the residual under-coverage is genuine distributional corruption. Split-conformal recalibration (2-fold series split, sort-before-calibrate) fixes that: c80 returns to 0.78–0.80 for every method at every bit-width, from RTN-W2-sim (0.060 $\rightarrow$ 0.782) to AWQ-W3 (0.480 $\rightarrow$ 0.795)—and raw fp32 itself under-covers (0.750; MACE 0.059 $\rightarrow$ 0.017 conformal), so a quantized-and-recalibrated model is better calibrated than the raw fp32 reference. Conformal retention is reported against the conformalized fp32 (same stage), leaving accuracy essentially unchanged. TimesFM-2.5 supplies independent evidence that this repair is a general TSFM need rather than a quantization artifact: with its auto-sort disabled, its fp32 model already emits crossed quantiles (QCR 0.109, c80 0.709)—the vendor ships exactly this sorting repair, on by default. Caveat: the conformal guarantee is series-level (residuals pooled across horizons), not per-horizon.

F8—the findings transfer across architectures. On encoder–decoder Chronos-Bolt small (48M) and base (205M), with the quantile head protected by default, *every* method is at parity: small W4-HQQ 0.994 and W3-GPTQ 0.986; base 0.9997–1.005 across all arms; calibration untouched (base c80 0.782 = fp32, QCR unchanged). On decoder-only TimesFM-2.5, likewise head-protected, W8-RTN 0.998, W4-GPTQ 0.999 [0.950, 1.034], and W4-HQQ 1.009 are at parity, while W3-GPTQ degrades (1.101 [1.054, 1.155])—its accuracy cliff sits one bit higher than Chronos-2’s, a genuine cross-model difference. Calibration is again untouched in every arm (c80 0.686–0.715 vs fp32’s 0.709; QCR 0.097–0.115 vs 0.109)—third-architecture confirmation of the head mechanism.

Table 2. Inference efficiency (storage-real variants, NVIDIA RTX 2000 Ada laptop, context 2048, horizon 64). Peak memory and latency at batch 1; throughput at batch 256.

Model	Size MB	Peak GPU MB	Lat. ms	Thr. ser./s
fp32	478	493	30	110
INT8 (HQQ)	120	176	42	122
NF4 (W4)	60	87	45	120

Efficiency. Storage-real 4-bit models occupy ≈ 60 MB ($8\times$ smaller) and cut peak batch-1 GPU memory from 493 MB (fp32) to 87 MB for NF4 ($5.7\times$; Table 2). Throughput is flat (~ 120 series/s at batch 256, vs 110 for fp32): our reference dequantize-then-GEMM paths buy footprint, not speed. Memory-bound and edge deployments benefit immediately.

4. Discussion, Limitations, Conclusion

Recommendation. (a) At 8 bits use any toolkit, but keep the 3-linear quantile head in fp32 (retention 0.9995 with fp32-level calibration). (b) For 4–3 bits use GPTQ or GPTAQ (GPTAQ-W3 1.016); below 3 bits PTQ collapses—use QAT. (c) Always sort predicted quantiles post-hoc: it is free, provably never worse, and zeroes quantile crossing. (d) When calibrated intervals matter, wrap the deployed model in split-conformal recalibration: it restores nominal coverage at any bit-width and even improves fp32. Report calibration and quantile-crossing alongside WQL whenever a probabilistic model is compressed.

Limitations. We study univariate zero-shot tasks; covariate/multivariate tasks and quantization-aware training (the indicated sub-3-bit path) are future work. Module-family sensitivity numbers are dev-tier (8 tasks). GPTQ/GPTAQ/AWQ arms are accuracy-exact simulations; realized speed needs packed kernels we do not implement (achievable footprints reported). Latency is hardware-specific; retention ratios are the transferable claim. Conformal guarantees are series-level, pooled across horizons. We assess but exclude KV-cache/vector methods such as TurboQuant (Zandieh et al., 2025), which do not apply to a cache-free encoder.

Conclusion. Across three architectures, TSFMs quantize remarkably well: 8-bit is free and error-compensating PTQ holds full-precision accuracy to 3–4 bits. The failure mode that does appear—calibration and monotonicity degrading before accuracy—is gated by 3.2% of the weights (the quantile head) and is preventable at the source (fp32 head) or curable post hoc (sorting + split-conformal) at any bit-width. Compressing a probabilistic forecaster is not merely an accuracy-retention problem—but with the head protected and quantiles recalibrated, it is close to a solved one down to 3 bits. Code and all per-series predictions are released.

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